

Equilibrium Information Acquisition in Optimal Auctions

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August 30, 2026

Abstract

We study non-cooperative information acquisition by bidders who will compete in an optimally-designed auction. The bidders simultaneously choose experiments privately revealing information about their respective willingnesses to pay. The auctioneer, knowing the experiments but not the realizations, designs an optimal auction against the resulting information structure. Bidders' best-responses are Myerson-regular: they require no ironing. In the symmetric equilibrium we characterize, each bidder adopts an information structure that satisfies a somewhat stronger regularity property. Among symmetric profiles with this property we show that it is the unique equilibrium.

Keywords: non-cooperative information design.

1 Introduction

We study equilibrium information acquisition by bidders in the face of a revenue-maximizing auctioneer. The bidders simultaneously choose costless experiments which privately reveal information about each's independent private value for a good to be sold at auction. The auctioneer sees the experiments but not the realization and then runs an auction that is designed optimally given the equilibrium information structure.

A starting point is [Roesler and Szentes \(2017\)](#) which can be thought of as a single-bidder version of our setting or, equivalently, the optimal collusive¹ arrangement for the two

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¹See also [Chen and Yang \(2023\)](#) who study the cooperative bidder-optimal *symmetric* information structure.

bidders. The consumer-surplus maximizing information structure yields efficient trade with probability 1 and has a distribution of values designed to induce the lowest-possible price. Inframarginally, bidders values are no smaller than necessary to dissuade the seller from sacrificing sales by raising price.

In our setting, in addition to moderating the surplus extraction of the auctioneer, each individual bidder's distribution of values dictates the terms of competition with the other bidder. We show as a consequence that trade occurs with a probability strictly less than 1. Indeed in equilibrium each bidder voluntarily places mass below the seller's (correctly anticipated) reserve price. For if bidder i exceeded the reserve for sure then bidder $-i$'s values just above the reserve price would be winning with probability zero. Bidder $-i$ would profitably deviate by "splitting" any mass there between a value of zero and higher values which earn rents. Inframarginal mass is constructed not just to generate rents but also to outcompete the opponent's inframarginal values. Thus [Roesler and Szentes's \(2017\)](#) iso-elastic demands are replaced by a condition on the *pass-through* schedule.

The symmetric equilibrium we analyze has other notable features. It has no "wasted" mass: all excluded types have value zero. Thus while the object sometimes goes unsold, in all such cases it has been revealed to be worthless to both bidders. The equilibrium information structure satisfies a strong form of [Myerson's \(1981\)](#) *regularity*: virtual value is monotone above the reserve and no ironing is needed in the design of the optimal auction.² The bidders' have positive mass at the maximum possible value and a smooth distribution elsewhere giving rise to a virtual value that ranges continuously from zero to 1. Indeed we show that it is the unique symmetric equilibrium in this class.

Our theoretical innovation is to develop tools for *non-cooperative* information design. Ours is a non-cooperative game in which two agents simultaneously choose information policies to influence the outcome of a downstream competition. Thus rather than a single optimal information design problem, our analysis characterizes the best-response *correspondence*: the information structure that maximizes bidder i 's payoff against a given information structure of the opponent. Our symmetric equilibrium information structure is a fixed point of that correspondence.

Related Literature [Persico \(2000\)](#) pioneered the study of information acquisition in auctions. That paper ranked (fixed) auction formats like first- and second-price auctions

²Naturally, we consider all possible deviations including value distributions that are not regular and require ironing. Our analysis thus includes "subgames" in which ironing is part of the optimal auction and we show these deviations are never profitable.

according to their risk-sensitivity and thus their incentives for costly information acquisition. [Kim and Koh \(2022\)](#) extended the analysis of fixed auction formats to allow for flexible costly information acquisition. [Bobkova \(2024\)](#), [Bergemann and Välimäki \(2002\)](#), and [Shi \(2012\)](#) also study various ways in which fixed-in-advance auction formats influence information acquisition incentives. When the auction format is fixed and when information design is flexible and costless it is always optimal to be fully informed.

Closest to our work are [Chen and Yang \(2023\)](#), [Bergemann et al. \(2024\)](#), and [Kyriazis \(2026\)](#). Both [Chen and Yang \(2023\)](#) and [Bergemann et al. \(2024\)](#) study cooperative buyer-optimal symmetric information acquisition before an optimal auction designer. The collective information structure is chosen to maximize total buyer surplus without the requirement that an individual bidder's information acquisition is a best-response. [Kyriazis \(2026\)](#) extends the [Roesler and Szentes \(2017\)](#) logic to a single-buyer non-linear screening environment in the framework of [Mussa and Rosen \(1978\)](#). As in our setting all of these papers have in common that Myerson-regular distributions feature in the solution. Also very close to our setting is [Terstiege and Wasser \(2022\)](#) who consider exogenously informed bidders who non-cooperatively disclose public information to the auctioneer before the design of an optimal auction.

Non-cooperative information design *per se* is not new, [Gentzkow and Kamenica \(2017\)](#) is an early contribution studying simultaneous persuasion mechanisms about a common state. [Boleslavsky and Cotton \(2018\)](#) and [Au and Kawai \(2020\)](#) both study a competitive persuasion game where competing proposers design information to persuade a receiver to adopt their own project. [Yang \(2015\)](#) analyzes equilibrium information acquisition about a common state before playing a coordination game.

2 Auction Preliminaries

We study an auction for an indivisible good with two competing bidders. Each bidder $i = 1, 2$ has a private value (or type) $v_i \in [0, 1]$ drawn from independent bidder-specific distribution functions F_i . The bidders are risk-neutral and their values are independent. The seller has zero opportunity cost of selling the object and, knowing the distributions F_i , designs a revenue-maximizing selling mechanism.

To describe the optimal auction we first represent the bidder's types and type distribu-

tions in terms of quantiles. For each $q_i \in [0, 1]$, define

$$V_i(q_i) = \sup \{v \in [0, 1] : \Pr_{F_i}\{v_i \geq v\} \geq q_i\}.$$

We can equivalently think of bidder i 's value being determined by first drawing a quantile q_i from the uniform distribution and then assigning value $V_i(q_i)$. We will maintain this interpretation throughout, effectively treating q_i as bidder i 's type. Note in particular that an atom in F_i at a value v_i corresponds to a flat segment of the value map $V_i(\cdot)$ and a flat segment in F_i corresponds to a discontinuous downward jump in V_i .

The map V_i is analogous to an inverse-demand function where $V_i(q_i)$ would be the largest price a monopolist could set and still sell a quantity (at least) q_i . Under this interpretation we may define an associated revenue function

$$R_i(q_i) = q_i V_i(q_i).$$

which will be useful in constructing the optimal auction for general (potentially badly-behaved) value distributions F_i .

We call the largest maximizer of a value map's revenue curve its *reserve quantile*:

$$Q = \max_{q \in [0, 1]} \arg \max R(q).$$

Thus, when several quantities attain the same monopoly revenue, the reserve quantile selects the largest quantity, or equivalently the lowest revenue-maximizing reserve price.

The Optimal Auction The revenue-maximizing auction is a *scoring-rule* auction in which values are assigned scores according to a non-increasing, bidder-specific function $\rho_i(q_i)$ and the object is assigned to the bidder with the highest score provided that score is non-negative. Ties are broken arbitrarily.

The specific scoring rule that defines the optimal auction is defined as follows. First, define the *ironed* revenue function

$$\bar{R}_i = \text{cav } R_i$$

as the pointwise-smallest concave function that is pointwise larger than R_i . The map $\bar{R}_i(q_i)$ is concave and with derivative

$$\bar{\rho}_i(q_i) = \frac{d}{dq_i} \bar{R}_i(q_i)$$

that exists at almost³ every quantile q_i .

As shown by Myerson (1981) and Bulow and Roberts (1989), an⁴ optimal auction uses the scoring rule

$$\rho_i(q_i) = \bar{\rho}_i(q_i)$$

3 Information Acquisition

We will consider the game in which the bidders first simultaneously acquire information about their values and then the auctioneer, seeing the information acquisition strategies but not their realizations, designs the optimal auction.

We suppose that the bidders have independent *ex post* valuations $\theta \in \{0, \bar{\theta}\}$ with $\bar{\theta}$ normalized to 1 and that begin with a (symmetric) prior expected valuation $m \in (0, 1)$. We model information acquisition as the flexible choice of an experiment leading to a private posterior expected value. Since the bidders are risk-neutral it is only this posterior expected value that matters for their auction-induced payoffs so we will equate it with v_i . Any such experiment can be described by its induced distribution over v_i . Indeed an experiment is feasible if and only if it induces a distribution over values v_i having mean m . Expressed in terms of quantiles a feasible experiment is equivalent to the choice of value map V_i satisfying

$$\int_0^1 V_i(q_i) dq_i = m.$$

Following Kamenica and Gentzkow (2011) we will refer to this *mean constraint* as Bayes' plausibility. Each bidder i 's pure strategy space is the set of Bayes' plausible value maps V_i .

Once the bidders have selected their strategies V_i , the seller designs an optimal incentive compatible and individually rational auction against (V_1, V_2) . The bidders then participate in the auction and each earns her corresponding payoff⁵ which we will denote by

$$\Pi_i(V_1, V_2).$$

³Because the ironed revenue curve is concave, it has left and right derivatives at every interior quantile. Throughout, at any kink we define ironed marginal revenue, and hence the bidder's score, using the left derivative. In particular, the score at a reserve quantile will be non-negative.

⁴The optimal auction is not unique. Ties in $\rho_i = \bar{\rho}_i$ can be broken arbitrarily with no effect on expected revenue.

⁵As noted above the optimal auction is not unique, and in particular for some choices of V_1, V_2 , the bidder's payoffs may depend on the particular optimal auction chosen. In equilibrium the bidders know in advance which choice would be made in each such a scenario and the payoffs incorporate this choice.

We will analyze symmetric pure-strategy equilibria of the simultaneous-move game played by the bidders, that is a value map V such that for all alternative pure strategies $\tilde{V}_i$

$$\Pi_i(V_1, V_2) \geq \Pi_i(\tilde{V}_i, V_{-i})$$

for $i = 1, 2$.

4 Regular Strategies and Interim Payoffs

Strategies that are *regular*⁶ play a special role. See [Figure 1](#) for illustration.

Definition 1. A Bayes' plausible value map V is *regular* if there exist $0 \leq p_1 < Q \leq 1$ such that $V(q) = 1$ for $q \leq p_1$, $V(q) = 0$ for $q > Q$ and over $q \in (p_1, Q)$, V is smooth and $V'(q) < 0$. Its marginal revenue

$$\rho(q) = V(q) + qV'(q)$$

is continuous and strictly decreasing on $[p_1, Q]$, with $\rho(Q) = 0$ and $\rho(p_1) = 1$.

We are going to show that there exists a symmetric equilibrium in regular strategies and in fact only one such equilibrium.⁷ To that end we will begin by characterizing best-responses to regular strategies.

If player $-i$ plays a regular strategy V_{-i} then the optimal-auction probability that player i wins with score $\rho_i \in [0, 1)$ is uniquely determined independent of the tie-breaking rule:

$$X_i(\rho_i) = 1 - \rho_{-i}^{-1}(\rho_i)$$

where $\rho_{-i}^{-1}(\rho_i)$ is the quantile for $-i$ earning score $\rho_i \in [0, 1)$.

Let $\rho_i(q_i)$ denote player i 's (possibly ironed) score and define

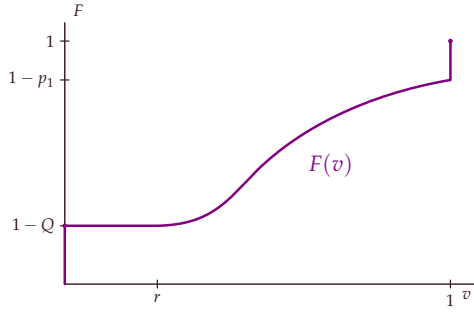
$$x_i(q_i) = X_i(\rho_i(q_i)).$$

When player i realizes the value at quantile q_i , her interim payoff is

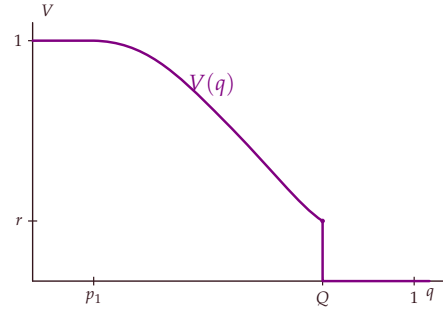
$$U(q_i) = V_i(q_i)x_i(q_i) - p(q_i).$$

⁶This definition is stronger than Myerson regularity which will be defined in the next section.

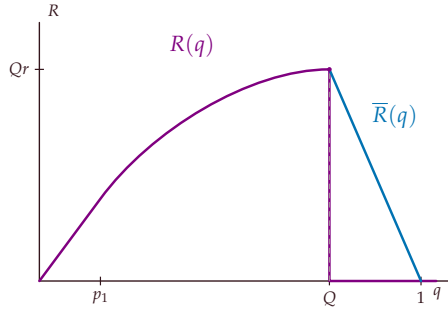
⁷Our arguments make the strong case that there can be no other symmetric equilibrium but we have not proven that yet.



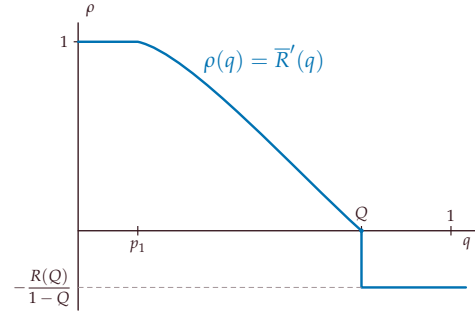
(a) Value distribution.



(b) Associated value map.



(c) Revenue and its concave envelope.



(d) Marginal revenue.

Figure 1: A regular value distribution and its auction representation. The atom at one becomes the flat segment $[0, p_1]$; the atom at zero becomes the flat segment $(Q, 1]$; and the support gap $(0, r)$ becomes the downward jump at the reserve quantile Q . Revenue is concave and nondecreasing through Q . Its concave envelope connects $(Q, R(Q))$ to $(1, 0)$, so marginal revenue equals one on the value-one atom, decreases to zero at the reserve quantile, and is negative on the value-zero atom.

By incentive compatibility,

$$U(q_i) = \max_{\hat{q}} \{V_i(q_i)x_i(\hat{q}) - p(\hat{q})\}.$$

Let us assume for the moment that bidder i uses a strategy V_i that is absolutely continuous.⁸ Then it is differentiable almost everywhere and by the envelope theorem,

$$U'(q_i) = V'_i(q_i)x_i(q_i) = V'_i(q_i)X_i(\rho_i(q_i))$$

⁸Absolute continuity of V_i allows for atoms in F_i but rules out gaps in the support of F_i as well as other singularities.

for almost every q_i . In an optimal auction the participation constraint binds, i.e.

$$U(1) = 0.$$

By the Fundamental Theorem of Calculus

$$\begin{aligned} U(q_i) &= U(1) - \int_{q_i}^1 U'(t) dt \\ &= - \int_{q_i}^1 X_i(\rho_i(t)) V_i'(t) dt. \end{aligned}$$

Quantiles are uniformly distributed, so player i 's ex ante payoff is

$$\Pi_i(V_i, V_{-i}) = \int_0^1 U(q_i) dq_i.$$

Substituting the interim-payoff formula gives

$$\Pi_i(V_i, V_{-i}) = - \int_0^1 \int_{q_i}^1 X_i(\rho_i(t)) V_i'(t) dt dq_i.$$

Changing the order of integration,

$$\begin{aligned} \Pi_i(V_i, V_{-i}) &= - \int_0^1 \left(\int_0^t dq_i \right) X_i(\rho_i(t)) V_i'(t) dt \\ &= - \int_0^1 t X_i(\rho_i(t)) V_i'(t) dt. \end{aligned}$$

Renaming t as q_i , we obtain

$$\Pi_i(V_i, V_{-i}) = - \int_0^1 q_i X_i(\rho_i(q_i)) V_i'(q_i) dq_i. \quad (1)$$

The integrand has a simple interpretation in terms of the information rents contributed at a quantile q_i . A declining value map $V_i'(q_i) < 0$ at quantile q_i generates rents for every lower quantile (i.e. higher value). Analogous to inverse demand, the slope represents a drop in the payment necessary to extend sales to lower quantiles. The rents from reduced payments accrue to all higher quantiles. The mass of those types is exactly q_i , and each receives the value decline multiplied by the allocation probability $X_i(\rho_i(q_i))$. Thus the

slope of the value map at q_i contributes

$$-q_i X_i(\rho_i(q_i)) V'_i(q_i)$$

to ex ante payoff. On an un-ironed portion of the revenue curve,

$$V_i(q_i) - \rho_i(q_i) = -q_i V'_i(q_i),$$

so the same contribution can be written as

$$X_i(\rho_i(q_i)) [V_i(q_i) - \rho_i(q_i)].$$

The slope of the value map therefore plays two roles: it creates information rents for higher types and, through marginal revenue, determines the score and hence the probability of allocation.

This derivation used the absolute continuity of V_i . An analogous representation holds for arbitrary value maps when the integral is interpreted in the Lebesgue–Stieltjes sense and the allocation at a downward jump is evaluated using the lower-value allocation.

Lemma 1 (Envelope representation). *Fix a regular strategy V_{-i} . For any Bayes' plausible value map V_i , player i 's optimal-auction payoff is*

$$\Pi_i(V_i, V_{-i}) = - \int_{(0,1]} q_i X_i(\rho_i(q_i)) dV_i(q_i)$$

where, at every downward jump of V_i , the allocation in the integrand is interpreted as the lower-value, right-hand allocation associated with the seller-optimal implementation.

The proof is in [appendix A](#).

5 Regularization

In this section we show one of the central ideas of the paper. Best-responses never entail ironing. Their implied value distribution always satisfy [Myerson's \(1981\)](#) regularity property.

Definition 2. A value map is Myerson-regular if its revenue curve is concave and nondecreasing on $[0, Q]$ and the value map is zero at every quantile above its reserve quantile Q .

We will show this by considering an arbitrary strategy V_i against a regular opponent V_{-i} . Suppose the revenue curve associated with V_i

$$R_i(q_i) = q_i V_i(q_i)$$

lies strictly below its concave envelope on an interval (a, b) . On this interval the concave envelope is a chord. Let s denote its slope, and suppose that $s > 0$.

The optimal auction assigns the constant score s to the entire interval $[a, b]$ and as a result the allocation probability

$$x = X_i(s).$$

is also constant.

We now construct the *regularized* strategy $\bar{V}_i$. Over the interval $[a, b]$, define

$$\bar{V}_i(q_i) = \frac{\bar{R}_i(q_i)}{q_i},$$

and set $\bar{V}_i(q_i) = V_i(q_i)$ elsewhere. Because the ironing chord lies above the raw revenue curve,

$$\bar{V}_i(q_i) \geq V_i(q_i) \quad \text{for } q_i \in [a, b].$$

The strategy $\bar{V}_i$ is not Bayes' plausible because it has a larger mean by an amount equal to

$$\Delta M = \int_a^b [\bar{V}_i(q_i) - V_i(q_i)] dq_i.$$

On the other hand, since $q_i \bar{V}_i(q_i) = \bar{R}_i(q_i)$ over the interval $[a, b]$, the regularized strategy has the same score and winning probability $x = X_i(s)$ over the affected interval. The payoff gain from regularization is thus

$$\begin{aligned} \Delta \Pi &= x \int_{[a,b]} q_i [dV_i(q_i) - d\bar{V}_i(q_i)] \\ &= x \int_a^b [\bar{V}_i(q_i) - V_i(q_i)] dq_i \\ &= x \Delta M. \end{aligned}$$

The second equality follows from Lebesgue–Stieltjes integration by parts. The boundary terms cancel because the raw and regularized revenue curves agree at the endpoints of the

ironing interval:

$$a\bar{V}_i(a) = aV_i(a) \quad \text{and} \quad b\bar{V}_i(b) = bV_i(b).$$

Thus regularizing the chord produces $X_i(s)$ units of additional payoff for every unit of mean it adds:

$$\Delta\Pi = X_i(s)\Delta M.$$

It is useful to organize this comparison by attaching a shadow price λ to mean and considering the dual objective

$$\Pi_i(V_i, V_{-i}) - \lambda M(V_i).$$

The multiplier measures the payoff cost of reducing the mean from changes elsewhere in the value map. One margin where it is always possible to do this is at the reserve Q_i . By marginally reducing bidder i 's reserve quantile without disturbing the value map above it, bidder i loses only against the reserve. This occurs only when the opponent's realized value comes from its atom at zero. The marginal payoff cost per unit of mean gained is thus

$$X_i(0) = 1 - Q_{-i},$$

suggesting that $\lambda = 1 - Q_{-i}$. The formal analysis confirms this and since

$$X_i(s) > X_i(0),$$

we can restrict attention to best-responses that are Myerson-regular.

6 Characterizing Best-Responses

In this section we will derive further necessary conditions for a best-response to a regular strategy. These conditions will then be used to obtain the paper's main result: derivation of the unique symmetric equilibrium in regular strategies.

The idea behind our characterization is to consider the cost and benefit to bidder i of increasing his value $V_i(q)$ at a quantile q . One cost comes from the increase in mean and we will capture that with a positive multiplier $\lambda > 0$ on the mean constraint. The direct benefit comes from increasing bidder i 's score at quantile q , which raises the probability of winning and generates rents for all lower quantiles. But there is a third, indirect effect that stems from the fact that raising $V_i(q)$ necessarily also affects the slope of V_i in the

neighborhood of q . Recall that the slope of V_i generates rents for bidder i and the third effect accounts for the resulting costs and benefits.

Following the developments in the previous section we start with a candidate best-response that is Myerson-regular. In particular its revenue function is concave and hence absolutely continuous up to its reserve quantile. At a point of differentiability q the score $\rho_i(q)$ is the marginal revenue

$$\rho_i(q) = V_i(q) + qV'_i(q)$$

By [Equation 1](#) the quantile q contributes

$$-qX_i(\rho_i(q))V'_i(q) \tag{2}$$

to i 's payoff. Marginally increasing $V_i(q)$ raises the score and increases i 's payoff by

$$B(q) = -qX'_i(\rho_i(q))V'_i(q).$$

This is the *level* effect. Next, the value map is required to be non-increasing so raising its level at q entails flattening its slope below q and steepening it above. For a $\hat{q}$ in the neighborhood, an increase in $V'_i(\hat{q})$ increases the score $\rho_i(\hat{q})$ by $\hat{q}$ thus adding $\hat{q}B(\hat{q})$ to the payoff. But it also has the direct effect on the integrand in [Equation 2](#) of $-\hat{q}x_i(\hat{q})$. The overall *slope* effect at $\hat{q}$ is

$$C(\hat{q}) = \hat{q}B(\hat{q}) - \hat{q}x_i(\hat{q})$$

This effect is added at quantiles below q where the slope must be raised, but it is subtracted at quantiles above. When the perturbation of V_i is concentrated at a small interval around quantile the first-order net payoff effect is $B(q) - C'(q)$ and at a best-response this cannot exceed the cost to the mean constraint:

$$B(q) - C'(q) \leq \lambda \tag{3}$$

These are local necessary conditions for a Myerson-regular best-response. When Myerson regularity is strengthened to full regularity, we are able to establish the following global properties of best-responses.

Proposition 1 (Regular best-responses). *A regular best-response to a regular strategy satisfies*

$$B(q) - C'(q) = \lambda$$

at every quantile q such that $V_i(q) \in (0, 1)$, where the multiplier on the mean constraint is

$$\lambda = 1 - Q_{-i}.$$

The proof is in [Appendix C](#). We can give a heuristic demonstration of some of the main structural features here. First, that the multiplier λ on the mean constraint must equal $1 - Q_{-i}$ follows from an argument analagous to the one given at the end of the previous section. Small variations in the reserve quantile exchange $1 - Q_{-i}$ payoff units per unit of mean.

Next, the inequality in [Equation 3](#) ensures that an *increase* in $V_i(q)$ cannot improve. Increases are always feasible but as V_i is necessarily weakly monotone, *decreases* are not feasible at flat segments of V_i . Nevertheless we can use the very same inequality to establish that a best-response must be strictly increasing on its interior range. As result two-sided local deviations are feasible and [Equation 3](#) must hold with equality.

To see this note that if $V_i(q)$ is locally constant then so is $X_i(\rho_i(q))$ and if $\rho_i(q) > 0$ then

$$B(q) - C'(q) = x_i(q) > X_i(0) = 1 - Q_{-i} = \lambda$$

violating [Equation 3](#). The same arugment shows that $\rho_i(\cdot)$ must be strictly decreasing wherever it is interior.

7 Constructing the Symmetric Equilibrium

Having identified the structure of best-responses, now symmetry adds the remaining structure necessary to identify a unique symmetric regular equilibrium candidate. When bidder i uses the same regular strategy as bidder $-i$, she wins exactly when she has the higher value, i.e. lower quantile.

$$X(\rho(q)) = 1 - q$$

for $q \in (p_1, Q]$. Thus $X'(\rho(q)) = -1/\rho'(q)$ and

$$B(q) = \frac{qV'(q)}{\rho'(q)} = q \frac{dV}{d\rho}.$$

The level effect becomes the classical pass-through rate $\frac{dV}{d\rho}$ of the inverse demand curve V .

Turning to the slope effect

$$C(q) = qB(q) - q(1 - q).$$

so that $C'(q) = B(q) + qB'(q) - (1 - 2q)$. Our necessary condition becomes

$$B(q) - C'(q) = -qB'(q) + 1 - 2q = \lambda.$$

Using $\lambda = 1 - Q$ and re-arranging,

$$B'(q) = \frac{Q}{q} - 2.$$

This last equation can be simply integrated to obtain the function B , i.e. the full pass-through schedule.

$$q \frac{dV}{d\rho} = 1 + Q - 2q + Q \log \left(\frac{q}{Q} \right). \quad (4)$$

It is known that the pass-through schedule identifies the inverse-demand curve up to an affine transformation (Weyl and Fabinger, 2013). Our normalization that marginal revenue has range $[0, 1]$ fixes its location and scale and therefore uniquely identifies the candidate equilibrium V .

All that remains is to pin down the sizes p_1 and $1 - Q$ of the atoms at values 1 and zero. First we must have $p_1 > 0$ because for any Q , the right-hand-side of Equation 4 eventually turns negative as $q \rightarrow 0$. By regularity, p_1 has score 1 and thus

$$1 = \rho(p_1) = V(p_1) + p_1 V'(p_1) = 1 + p_1 V'(p_1)$$

implying that $V'(p_1) = 0$. There is smooth pasting of the value map at p_1 . In particular the pass-through is zero at p_1 and thus

$$B(p_1) = 0$$

defines $p_1 > 0$ and the entire family of solutions to Equation 4 is indexed by the scalar

parameter Q . Letting V_Q denote the solution for a given Q , we can calculate its mean,

$$M(V_Q) = \int_0^1 V_Q(q) dq.$$

We show that M is continuous, strictly increasing and onto $(0, 1)$, thus the mean constraint

$$M(V_Q) = m.$$

determines Q and the candidate equilibrium. Because a best-response exists and must satisfy the necessary conditions in [Proposition 1](#), we can show that the resulting strategy is a best-response to itself and hence an equilibrium, indeed the unique symmetric regular equilibrium.

Theorem 1. *The unique symmetric regular equilibrium is the Bayes-plausible value map V^* satisfying*

$$q \frac{dV^*}{d\rho^*} = 1 + Q^* - 2q + Q^* \log \left(\frac{q}{Q^*} \right)$$

for all $q \in [p_1^*, Q^*]$, where p_1^* and Q^* are defined by

$$\rho^*(p_1^*) = 1 \quad \text{and} \quad \rho^*(Q^*) = 0.$$

[Figure 2](#) illustrates the equilibrium for three prior means. As the mean increases, the atom at one grows while the atom at zero shrinks and the smooth section adjusts.

8 Going Forward

Our analysis would seem to extend directly to the case of many bidders. When bidders have exogenous initial private information the information design is constrained to be no less informative than that prior. It would be natural to consider the extension in which that constraint is imposed ([Chopra and Ely, 2025](#)). Indeed in that case an external information provider would have to elicit the initial private information of the bidder in order to provide tailored information in an incentive compatible way motivating a further variation in which these incentive constraints are imposed ([Chopra and Ely, 2026](#)).

Moving beyond information design, one can interpret a bidder's distribution of values as arising from an *ex ante* investment stage as in [Condorelli and Szentes \(2020\)](#). Formally this interpretation removes the Bayes' plausibility constraint. Interestingly, unlike in

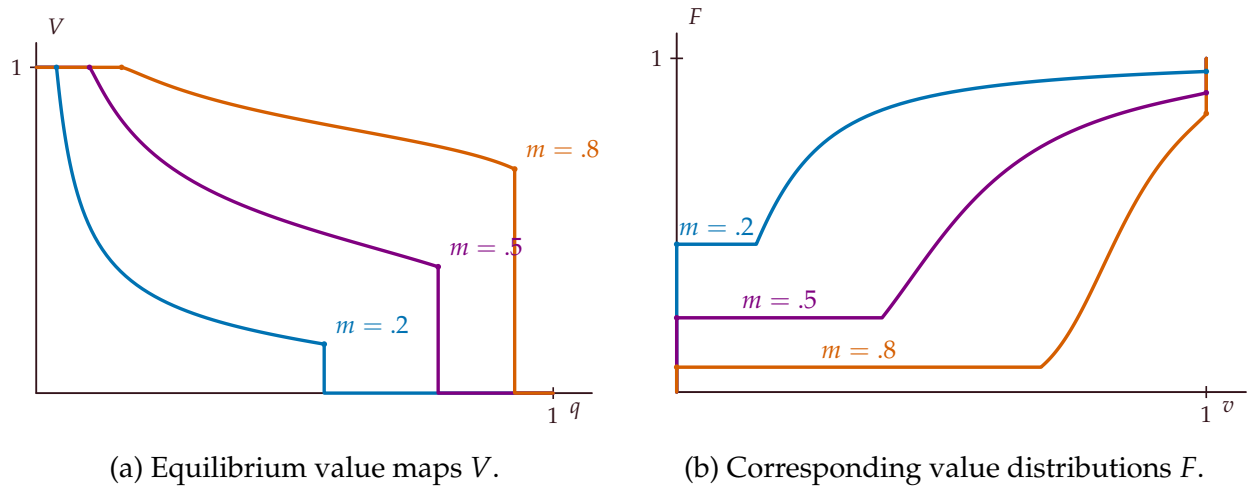


Figure 2: The symmetric regular equilibrium for prior means $m = .2, .5, .8$. The curves are numerical solutions of the pass-through equation in [Theorem 1](#). The right panel shows the associated CDFs.

the single-buyer setting of [Condorelli and Szentes \(2020\)](#) and [Roesler and Szentes \(2017\)](#) where the investment/information design solutions closely resemble one another, with competing bidders the framework of costless investment leads to a degenerate solution akin to Bertrand competition: the unique equilibrium has both bidders placing all mass on their highest possible value.⁹ Adding investment costs would restore a non-trivial tradeoff and lead to an interesting extension.

⁹It is easy to see that such a profile is an equilibrium: any deviation to lower value would simply lose the auction to the opponent's atom at value 1. The uniqueness argument is only slightly more involved.

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Appendix

A Envelope Representation

Proof of lemma 1. We pick up from the quantile-space derivation in the text. It remains to extend that argument from absolutely continuous to arbitrary value maps. Write

$$U(q_i) = V_i(q_i)x_i(q_i) - p(q_i).$$

For any q_i and $\hat{q}$ such that $q_i < \hat{q}$, incentive compatibility gives

$$x_i(\hat{q})[V_i(q_i) - V_i(\hat{q})] \leq U(q_i) - U(\hat{q}) \leq x_i(q_i)[V_i(q_i) - V_i(\hat{q})].$$

Summing these inequalities over partitions gives lower and upper Riemann–Stieltjes sums. On every continuous portion of the value map, their common limit is the quantile-space envelope formula used in the text.

It remains to identify the contribution of a downward jump. Suppose that at quantile q_i the value map jumps from $\bar{v}$ to $\underline{v}$, and denote the allocations assigned to those two values by $\bar{x}$ and $\underline{x}$, respectively. The preceding incentive constraints become

$$\underline{x}(\bar{v} - \underline{v}) \leq U(\bar{v}) - U(\underline{v}) \leq \bar{x}(\bar{v} - \underline{v}).$$

For a fixed allocation rule, the seller-optimal implementation minimizes bidders' utilities subject to incentive compatibility and the normalization of the lowest type's utility. It therefore selects the lower bound: the utility increment across the gap is

$$\underline{x}(\bar{v} - \underline{v}).$$

The mass of types above the gap is q_i , so this jump contributes

$$q_i \underline{x}(\bar{v} - \underline{v})$$

to ex ante utility. This is precisely the atom of the Lebesgue–Stieltjes integral when its allocation is evaluated at the lower value. Combining the continuous and jump components, and using

$$x_i(q_i) = X_i(\rho_i(q_i)),$$

gives the stated representation. □

B Dual Regularization

Fix a regular opponent and, as in the regularization section, set

$$\lambda = X_i(0) = 1 - Q_{-i}.$$

For a decreasing value map, not necessarily satisfying the mean constraint, write

$$\mathcal{D}_i(V_i; V_{-i}) = \Pi_i(V_i, V_{-i}) - \lambda M(V_i).$$

Proposition 2 (Dual regularization). *For every decreasing value map V_i , there is a Myerson-regular map $\widehat{V}_i$ such that*

$$\mathcal{D}_i(\widehat{V}_i; V_{-i}) \geq \mathcal{D}_i(V_i; V_{-i}).$$

Proof. Let Q_i be the reserve quantile of V_i . The concavification $\bar{R}_i$ has the same largest maximizer as R_i , so Q_i is also the end of the region on which $\bar{R}_i$ has nonnegative slope. Define

$$\widehat{V}_i(q_i) = \begin{cases} \bar{R}_i(q_i)/q_i, & 0 < q_i \leq Q_i, \\ 0, & Q_i < q_i \leq 1, \end{cases}$$

with its value at zero defined by the right limit. Concavity of $\bar{R}_i$ and $\bar{R}_i(0) = 0$ imply that $\bar{R}_i(q_i)/q_i$ is decreasing. Because values lie in $[0, 1]$, we have $\bar{R}_i(q_i) \leq q_i$, and therefore $0 \leq \widehat{V}_i \leq 1$. The revenue curve of $\widehat{V}_i$ agrees with $\bar{R}_i$ through Q_i and is zero thereafter. Its concave envelope joins $(Q_i, \bar{R}_i(Q_i))$ to $(1, 0)$. Thus $\widehat{V}_i$ is Myerson-regular.

On each component over which $\bar{R}_i$ strictly exceeds R_i , the calculation in [Section 5](#) applies with constant score $\bar{\rho}_i$. Summing that calculation over the ironing components gives

$$\begin{aligned} & \mathcal{D}_i(\widehat{V}_i; V_{-i}) - \mathcal{D}_i(V_i; V_{-i}) \\ &= \int_0^{Q_i} [X_i(\bar{\rho}_i(q_i)) - X_i(0)][\widehat{V}_i(q_i) - V_i(q_i)] dq_i + X_i(0) \int_{Q_i}^1 V_i(q_i) dq_i. \end{aligned}$$

Both terms are nonnegative. On the first interval, $\widehat{V}_i \geq V_i$ and $\bar{\rho}_i \geq 0$. Beyond Q_i , the original negative-score types are still excluded. □

C Regular Best Responses

We now prove the characterization in [Proposition 1](#).

Proof of Proposition 1. The scalar mean constraint has a multiplier λ . On the smooth interior of a regular V_i , write

$$\rho_i(q_i) = V_i(q_i) + q_i V_i'(q_i).$$

For a smooth, compactly supported perturbation of V_i , direct differentiation of the payoff integrand gives

$$\delta \Pi_i = \int [B(q_i) \delta V_i(q_i) + C(q_i) \delta V_i'(q_i)] dq_i,$$

where

$$B(q_i) = -q_i X_i'(\rho_i(q_i)) V_i'(q_i)$$

and

$$C(q_i) = q_i B(q_i) - q_i X_i(\rho_i(q_i)).$$

Because both signs of every compactly supported interior perturbation are feasible, integration by parts and the fundamental lemma of the calculus of variations give

$$B(q_i) - C'(q_i) = \lambda$$

for every q_i such that $V_i(q_i) \in (0, 1)$.

It remains to identify the multiplier. Let Q_i be the reserve quantile of V_i and let $r_i = V_i(Q_i) > 0$. By regularity the score at the reserve quantile is zero, i.e. $\rho_i(Q_i) = 0$ hence $r_i = -Q_i V_i'(Q_i)$. The value $v_i(Q_i)$ and the reserve location Q_i are free subject to these conditions. Their two transversality conditions are

$$C(Q_i) = 0 \quad \text{and} \quad r_i [B(Q_i) - \lambda] = 0.$$

The first implies

$$B(Q_i) = X_i(0),$$

and the second then implies

$$\lambda = X_i(0) = 1 - Q_{-i}.$$

□

D Constructing the Symmetric Equilibrium Candidate

Suppose both bidders use the same regular value map. For $q \in (p_1, Q)$, symmetry gives

$$X(\rho(q)) = 1 - q, \quad X'(\rho(q)) = -\frac{1}{\rho'(q)}.$$

It follows that

$$B(q) = \frac{qV'(q)}{\rho'(q)} = q \frac{dV}{d\rho}$$

and

$$C(q) = qB(q) - q(1 - q).$$

Substituting into the necessary conditions in [Proposition 1](#), we have

$$-qB'(q) + 1 - 2q = 1 - Q = B(Q).$$

For any value of Q , integration yields the candidate function $B_Q(q)$

$$B_Q(q) = 1 + Q - 2q + Q \log \left(\frac{q}{Q} \right). \quad (5)$$

It must satisfy the boundary conditions

$$V_Q(p_1(Q)) = 1, \quad \rho_Q(Q) = 0. \quad (6)$$

Lemma 2. *For each $Q \in (0, 1)$, there is a unique regular value map V_Q that satisfies the pass-through equation in [Equation 5](#) and the boundary conditions in [Equation 6](#). It has a value-1 atom $p_1(Q)$ given by the unique root in $(0, Q/2)$ of*

$$1 + Q - 2p_1(Q) + Q \log \left(\frac{p_1(Q)}{Q} \right) = 0.$$

Proof. The right side of [Equation 5](#) tends to $-\infty$ as q decreases to zero and has derivative $Q/q - 2$. It is therefore strictly increasing on $(0, Q/2)$ and has a unique root there.

Define a function g_Q by

$$\frac{g'_Q(q)}{g_Q(q)} = \frac{1}{B_Q(q)} - \frac{2}{q}.$$

with $g_Q(Q) = 1$. Every solution to the pass-through equation has

$$V'_Q(q) = -kg_Q(q)$$

for some $k > 0$. Because $B_Q(p_1(Q)) = 0$ and $B'_Q(p_1(Q)) > 0$, the differential equation implies that $g_Q(q)$ converges to zero as q decreases to $p_1(Q)$. Since V_Q is constant on $[0, p_1(Q)]$, any solution beginning at $p_1(Q)$ is differentiable there, with $V'_Q(p_1(Q)) = 0$, and therefore $\rho_Q(p_1(Q)) = 1$, as required by regularity.

Define

$$G_Q(q) = \int_{p_1(Q)}^q g_Q(t) dt.$$

Using the condition $V_Q(p_1(Q)) = 1$ in [Equation 6](#), integration gives

$$V_Q(q) = 1 - kG_Q(q).$$

Because $g_Q(Q) = 1$, the condition $\rho_Q(Q) = 0$ in [Equation 6](#) becomes

$$0 = 1 - kG_Q(Q) - kQ.$$

Thus

$$V_Q(q) = 1 - \frac{G_Q(q)}{Q + G_Q(Q)}$$

on $[p_1(Q), Q]$, with value one on $[0, p_1(Q)]$ and value zero on $(Q, 1]$. This also proves uniqueness. $\square$

The mean of V_Q is

$$M(Q) = Q - \frac{J_Q}{Q + G_Q(Q)} \tag{7}$$

where

$$J_Q = \int_{p_1(Q)}^Q G_Q(q) dq.$$

Lemma 3. *The map $Q \mapsto M(Q)$ is continuous and strictly increasing from zero to one. Hence each $m \in (0, 1)$ identifies a unique Q^* for which $M(Q) = m$.*

Proof. Write $q = Qt$ and define

$$b(t) = 1 - 2t + \log t.$$

Then

$$B_Q(Qt) = 1 + Qb(t),$$

and $t_0(Q)$ is determined by

$$1 + Qb(t_0(Q)) = 0.$$

Because b is strictly increasing on $(0, 1/2)$, $t_0(Q)$ is continuous and strictly increasing in Q .

Define

$$\gamma_Q(t) = g_Q(Qt)$$

on $[t_0(Q), 1]$, and set it equal to zero below $t_0(Q)$. It satisfies

$$\frac{\gamma'_Q(t)}{\gamma_Q(t)} = \frac{Q}{1 + Qb(t)} - \frac{2}{t}, \quad \gamma_Q(1) = 1.$$

Changing variables and reversing the order of integration in [Equation 7](#) gives

$$G_Q(Q) = Q \int_0^1 \gamma_Q(t) dt$$

and

$$J_Q = Q^2 \int_0^1 (1-t) \gamma_Q(t) dt.$$

Consequently,

$$M(Q) = Q \frac{1 + \int_0^1 t \gamma_Q(t) dt}{1 + \int_0^1 \gamma_Q(t) dt}.$$

Define the probability measure ν_Q by

$$d\nu_Q(t) = \frac{\delta_1(dt) + \gamma_Q(t) dt}{1 + \int_0^1 \gamma_Q(u) du}.$$

The preceding expression for the mean is simply

$$M(Q) = Q \int_0^1 t d\nu_Q(t).$$

Now fix $0 < Q_1 < Q_2$. On the common support of the two densities,

$$\frac{d}{dt} \log \frac{\gamma_{Q_2}(t)}{\gamma_{Q_1}(t)} = \frac{Q_2 - Q_1}{[1 + Q_2 b(t)][1 + Q_1 b(t)]} > 0.$$

The unnormalized ratio equals one at $t = 1$. Relative to ν_{Q_1} , its normalized version is zero on $[t_0(Q_1), t_0(Q_2))$, proportional to

$$\frac{\gamma_{Q_2}(t)}{\gamma_{Q_1}(t)}$$

on $[t_0(Q_2), 1)$, and takes the limiting proportionality constant at the atom at one. Denote this normalized ratio by r . It is nondecreasing and nonconstant, and

$$\int_0^1 r(t) d\nu_{Q_1}(t) = 1.$$

Since both t and $r(t)$ are increasing,

$$\begin{aligned} \int_0^1 t d\nu_{Q_2} - \int_0^1 t d\nu_{Q_1} &= \text{Cov}_{\nu_{Q_1}}(t, r(t)) \\ &> 0. \end{aligned}$$

It follows that both factors in

$$M(Q) = Q \int_0^1 t d\nu_Q(t)$$

are strictly increasing in Q .

Continuity follows from continuous dependence of γ_Q on Q and dominated convergence. Because ν_Q is supported on $[0, 1]$,

$$0 < M(Q) \leq Q,$$

so $M(Q)$ converges to zero with Q . Finally, for each fixed $t < 1$,

$$\log \gamma_Q(t) = - \int_t^1 \left[\frac{Q}{1 + Qb(u)} - \frac{2}{u} \right] du.$$

The integral of the first term diverges as Q increases to one, while the integral of the second remains finite. Hence $\gamma_Q(t)$ converges to zero for every $t < 1$. The preceding likelihood-ratio comparison also shows that $\gamma_Q(t)$ decreases with Q for every $t < 1$, so any fixed γ_Q supplies an integrable dominator for all larger values of Q . Dominated convergence then gives $\nu_Q \Rightarrow \delta_1$, and therefore $M(Q)$ converges to one. $\square$

E Ruling Out Deviations

We now consider all possible deviations, regular and otherwise, from the candidate equilibrium profile. Set

$$\lambda = 1 - Q^*, \quad r^* = V^*(Q^*).$$

Let X^* be the allocation probability induced by the candidate opponent. For $x \in (p_1^*, Q^*)$, Equation 5 gives

$$B_{Q^*}(x) = x \frac{dV^*(x)}{d\rho^*(x)}.$$

We will make use of the following implications of symmetry and the pass-through equation Equation 5:

$$X^*(\rho^*(x)) = 1 - x, \tag{8}$$

$$\frac{dX^*(\rho^*(x))}{d\rho} = - \left(\frac{d\rho^*(x)}{dx} \right)^{-1}, \tag{9}$$

$$V^*(x) - \rho^*(x) = -B_{Q^*}(x) \frac{d\rho^*(x)}{dx}, \tag{10}$$

$$\frac{d}{dx}[x + B_{Q^*}(x)] = \frac{Q^*}{x} - 1 > 0, \quad Q^* + B_{Q^*}(Q^*) = 1.$$

Lemma 4. *There is a continuously differentiable function h on $[0, 1]$, taking nonpositive values, such that $h(v) = 0$ for $v \leq r^*$ and*

$$X^*(\rho)(v - \rho) - \lambda v \leq h(v) + h'(v)(\rho - v) \tag{11}$$

whenever $0 \leq \rho \leq v \leq 1$. Equality holds at every value–score pair used by the nonnegative-score portion of V^* .

Proof. The value map V^* decreases strictly from 1 to r^* over the interval $q \in (p_1^*, Q^*)$. For $v \in [r^*, 1]$, let $x(v)$ be its inverse and define

$$h(v) = \frac{dX^*(\rho^*(x(v)))}{d\rho} [v - \rho^*(x(v))]^2 - \lambda v. \tag{12}$$

Set $h(v) = 0$ for $v \leq r^*$.

We first calculate its derivative. To ease notation write

$$u = V^* - \rho^*, \quad p = \frac{dX^*(\rho^*)}{d\rho} u - X^*(\rho^*).$$

Using Equation 8–Equation 10,

$$p = x + B_{Q^*}(x) - 1, \quad \frac{dp}{dx} = \frac{X^*(\rho^*(x)) - \lambda}{x}.$$

Using $u = -x \frac{dV^*(x)}{dx}$, the derivative of h evaluated at $V^*(x)$ is

$$h'(V^*(x)) = p = x + B_{Q^*}(x) - 1. \quad (13)$$

Together, Equation 12 and Equation 13 show that Equation 11 holds with equality at every value–score pair $(V^*(x), \rho^*(x))$ with $x \in (0, 1)$. At the reserve, Equation 9 and Equation 10 give

$$\frac{dX^*(0)}{d\rho} r^* = B_{Q^*}(Q^*) = \lambda.$$

Hence

$$h(r^*) = 0, \quad h'(r^*) = 0.$$

The function h is therefore constant below r^* , hence continuously differentiable there. Since $x + B_{Q^*}(x)$ is increasing and equals one at Q^* , Equation 13 also implies $h \leq 0$. Moreover, $h(1) = -\lambda$ and

$$\lim_{v \uparrow 1} h'(v) = \lim_{x \downarrow p_1^*} [x + B_{Q^*}(x) - 1] = p_1^* - 1.$$

Thus h is continuously differentiable on all of $[0, 1]$.

It remains to prove the inequality in Equation 11. Begin with $v \in (r^*, 1)$ and let $x = x(v)$. Define

$$\psi(q) = [X^*(\rho^*(q)) + h'(v)][v - \rho^*(q)].$$

We are going to show that $q = x$ uniquely maximizes ψ . Since for every $\hat{\rho} \in (0, 1)$ there exists a q such that $\rho^*(q) = \hat{\rho}$, it will follow that $\rho^*(x)$ uniquely maximizes

$$[X^*(\rho) + h'(v)](v - \rho)$$

over all interior scores ρ , and therefore that the inequality in Equation 11 holds for this

value v and every interior score. Direct differentiation gives

$$\begin{aligned}\psi'(q) &= \rho^*(q) - [x + B_{Q^*}(x) - q] \frac{d\rho^*(q)}{dq} - V^*(x) \\ &= V^*(q) - V^*(x) - [x + B_{Q^*}(x) - q - B_{Q^*}(q)] \frac{d\rho^*(q)}{dq}.\end{aligned}$$

Because V^* is strictly decreasing, $q + B_{Q^*}(q)$ is strictly increasing, and $\frac{d\rho^*(q)}{dq} < 0$,

$$\psi'(q) > 0 \quad \text{if } q < x, \quad \psi'(q) < 0 \quad \text{if } q > x.$$

Thus $q = x$ uniquely maximizes ψ . At that quantile,

$$\psi(x) = \frac{dX^*(\rho^*(x))}{d\rho} [v - \rho^*(x)]^2 = h(v) + \lambda v.$$

Rearranging the resulting maximization inequality proves [Equation 11](#). The preceding argument covers scores in $(0, 1)$. At score zero, the inequality follows by taking ρ down to zero and using $X^*(0) = \lim_{\rho \downarrow 0} X^*(\rho)$. At score one, feasibility implies $v = 1$, so $v - \rho = 0$, and the inequality follows from $h(1) = -\lambda$.

Finally, suppose $v \leq r^*$. By continuity, the inequality also holds at r^* . Since both h and its derivative vanish there, it gives

$$X^*(\rho)(r^* - \rho) - \lambda r^* \leq 0.$$

For $0 \leq \rho \leq v \leq r^*$, lowering value from r^* changes the left side by

$$[X^*(\rho) - \lambda](v - r^*) \leq 0,$$

because $X^*(\rho) \geq X^*(0) = \lambda$. This proves the inequality on the remaining value range. $\square$

Proof of [Theorem 1](#). First let V_i be Myerson-regular, with reserve quantile Q_i , and let $R_i(q_i)$ be its revenue function. Set $\rho_i = R'_i$ on $[0, Q_i]$. Concavity and $R_i(0) = 0$ imply, almost everywhere,

$$0 \leq \rho_i(q_i) \leq V_i(q_i), \quad q_i V'_i(q_i) = \rho_i(q_i) - V_i(q_i).$$

The value map V_i is absolutely continuous on $[0, Q_i]$. [Lemma 1](#) and the identity above give

$$\Pi_i(V_i, V^*) = - \int_0^{Q_i} q_i X^*(\rho_i(q_i)) V_i'(q_i) dq_i = \int_0^{Q_i} X^*(\rho_i(q_i)) [V_i(q_i) - \rho_i(q_i)] dq_i.$$

We can apply [Lemma 4](#).

$$\begin{aligned} \mathcal{D}_i(V_i; V^*) &= \int_0^{Q_i} \{X^*(\rho_i(q_i)) [V_i(q_i) - \rho_i(q_i)] - \lambda V_i(q_i)\} dq_i \\ &\leq \int_0^{Q_i} \{h(V_i(q_i)) + h'(V_i(q_i)) [\rho_i(q_i) - V_i(q_i)]\} dq_i \\ &= \int_0^{Q_i} \{h(V_i(q_i)) + q_i h'(V_i(q_i)) V_i'(q_i)\} dq_i \\ &= \int_0^{Q_i} \frac{d}{dq_i} \{q_i h(V_i(q_i))\} dq_i \\ &= Q_i h(V_i(Q_i)) \leq 0. \end{aligned}$$

The last equality is the fundamental theorem of calculus. The first inequality follows from [Lemma 4](#). In the case of $V_i = V^*$, that inequality is an equality and also $h(V^*(Q^*)) = h(r^*) = 0$. Therefore

$$\mathcal{D}_i(V^*; V^*) = 0.$$

Now let V_i be an arbitrary decreasing value map. [Proposition 2](#) supplies a Myerson-regular map $\widehat{V}_i$ such that

$$\mathcal{D}_i(V_i; V^*) \leq \mathcal{D}_i(\widehat{V}_i; V^*) \leq 0.$$

Every Bayes' plausible deviation has the same mean m as V^* . Hence

$$\Pi_i(V_i, V^*) - \Pi_i(V^*, V^*) = \mathcal{D}_i(V_i; V^*) - \mathcal{D}_i(V^*; V^*) \leq 0.$$

Thus V^* is a best response to itself.

Finally, any symmetric regular equilibrium must satisfy [Proposition 1](#). Symmetry then gives [Equation 5](#). [Lemma 2](#) identifies the unique regular map for each reserve quantile and [Lemma 3](#) identifies the unique reserve quantile compatible with Bayes' plausibility. Thus V^* is the unique symmetric regular equilibrium. $\square$